

House Price Prediction using Linear Regression

Artificial Intelligence & Machine Learning Internship Project Report

1. Introduction

Machine Learning is one of the most rapidly growing fields in Artificial Intelligence. It enables systems to learn patterns from data and make predictions without being explicitly programmed. In this project, a Machine Learning-based Linear Regression model was developed to predict house prices using the California Housing dataset. The project demonstrates the complete Machine Learning workflow including data loading, exploratory data analysis (EDA), preprocessing, model training, prediction, evaluation, and visualization using Python and scikit-learn libraries.

2. Objective

The main objective of this project is to build and evaluate a house price prediction model using the Linear Regression algorithm. The project aims to understand the practical implementation of Machine Learning concepts such as feature analysis, model training, evaluation metrics, and data visualization. Another important objective is to gain hands-on experience with Python libraries such as pandas, NumPy, matplotlib, seaborn, and scikit-learn.

3. Dataset Description

The California Housing dataset from scikit-learn was used in this project. The dataset contains various housing-related numerical features including: median income, house age, average rooms, population, latitude, and longitude. These features are used to predict the target variable, which represents the median house value. The dataset is widely used for regression-based Machine Learning projects and educational purposes.

4. Methodology

The methodology adopted in this project consists of multiple stages. First, the dataset was loaded and converted into a pandas DataFrame. Exploratory Data Analysis (EDA) was performed to understand feature distributions, correlations, and missing values. After preprocessing, the dataset was divided into training and testing sets using `train_test_split()`. A Linear Regression model was then trained using the training dataset. Predictions were generated using the testing dataset and evaluated using MAE, RMSE, and R^2 Score metrics. Finally, different visualizations such as correlation heatmaps, prediction scatter plots, and residual plots were generated for better understanding of model performance.

5. Dataset Sample

MedInc	HouseAge	AveRooms	Population	Latitude	Longitude	target
4.0	9.0	6.1	3159.0	33.13	-120.07	3.87

3.36	37.0	4.73	4648.0	41.31	-115.71	3.83
4.15	33.0	5.78	3059.0	41.74	-118.31	4.69
5.02	42.0	6.32	2365.0	41.96	-123.36	4.89
3.27	44.0	2.78	2579.0	32.56	-123.63	4.82

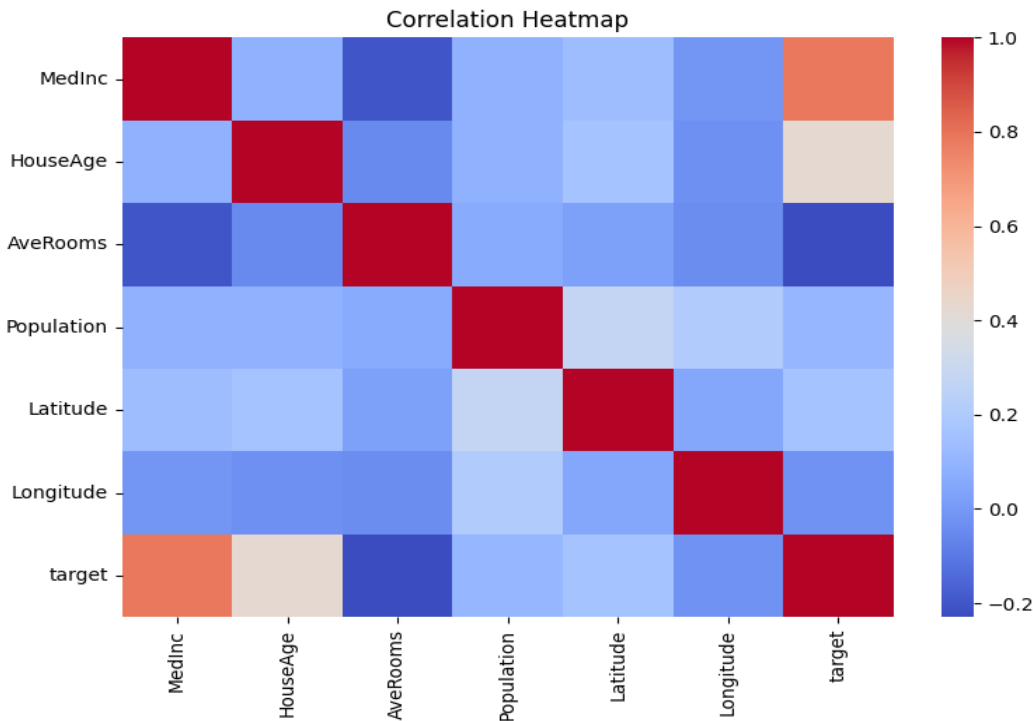
6. Model Evaluation

The performance of the Linear Regression model was evaluated using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score. Lower MAE and RMSE values indicate better prediction accuracy, while a higher R² Score indicates stronger explanatory power of the model.

Metric	Value	Description
MAE	0.5332	Average prediction error
RMSE	0.7456	Measures prediction deviation
R ² Score	0.5758	Explains variance captured by model

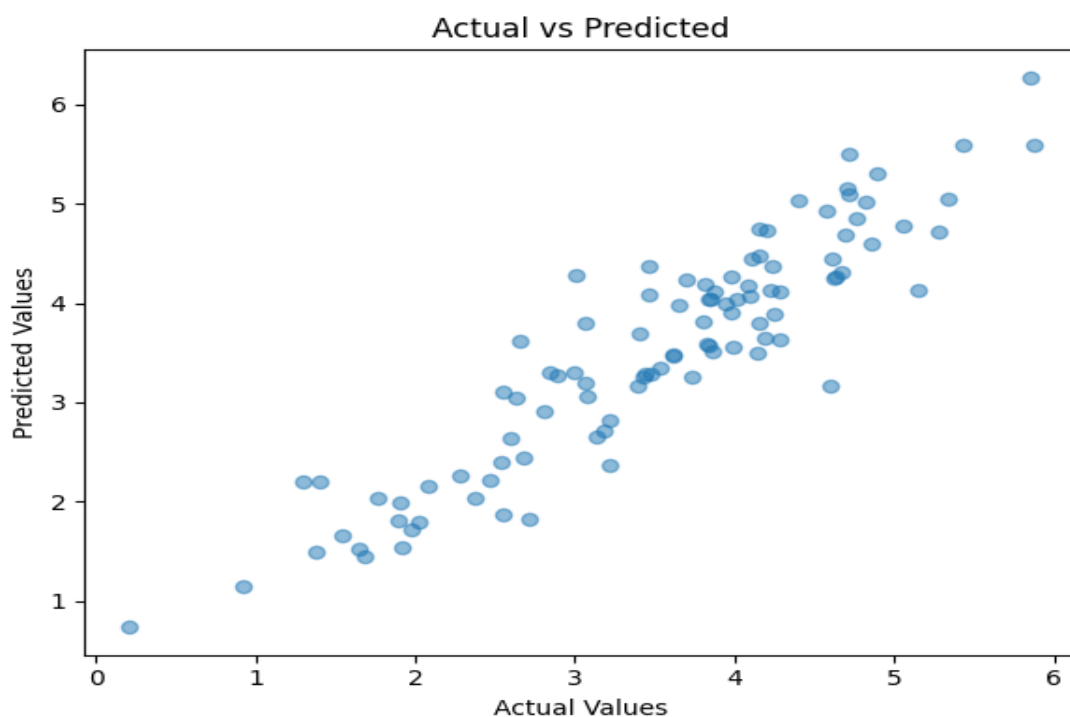
7. Correlation Heatmap

The heatmap visualizes relationships between different numerical features in the dataset. Stronger colors represent stronger correlations between variables.



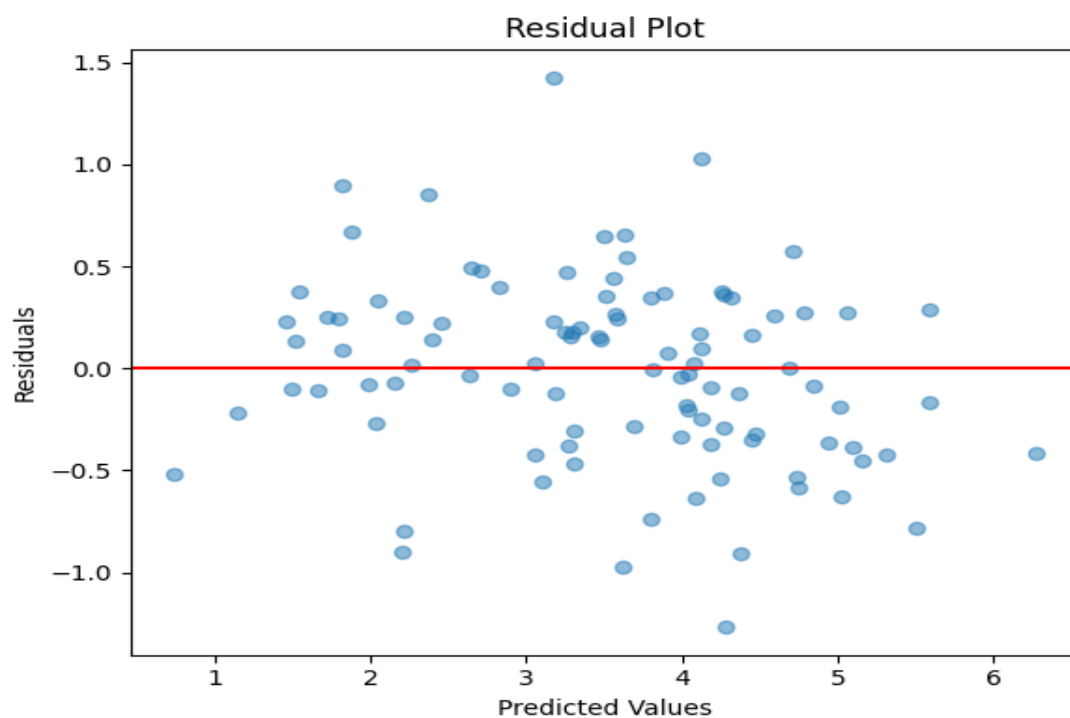
8. Actual vs Predicted Plot

This scatter plot compares actual house prices with predicted values generated by the Linear Regression model. Closer alignment indicates better prediction accuracy.



9. Residual Plot

The residual plot visualizes prediction errors. Ideally, residuals should be randomly distributed around zero, indicating good model behavior.



10. Conclusion

The project successfully implemented a Machine Learning-based house price prediction system using the Linear Regression algorithm. The project provided practical understanding of the complete Machine Learning lifecycle including EDA, preprocessing, training, evaluation, and visualization. The model achieved reasonable predictive performance and demonstrated how regression techniques can be applied to real-world datasets. The project also enhanced practical skills in Python, data analysis, and Machine Learning libraries.

11. Future Improvements

Future improvements for this project may include:

- Feature scaling and normalization
- Hyperparameter tuning
- Using advanced algorithms such as Random Forest and XGBoost
- Model deployment using Flask or Streamlit
- Real-time prediction interface development